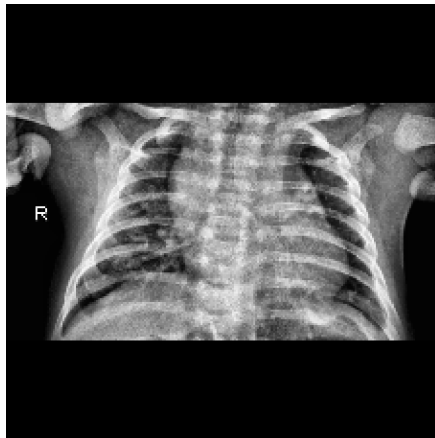


Figure 8. Grad-CAM visualization of image regions contributing to a representative Proposed HydroFed-ICAF prediction.

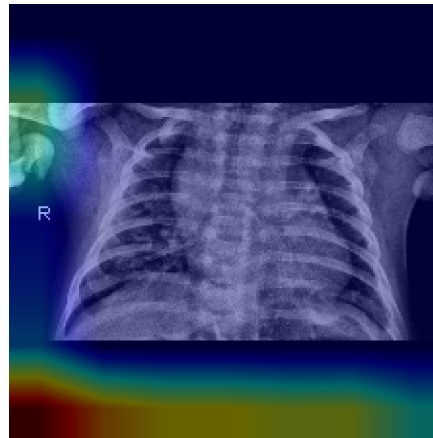
**(a) Original Chest X-Ray
(Pediatric Patient 10-V656)**



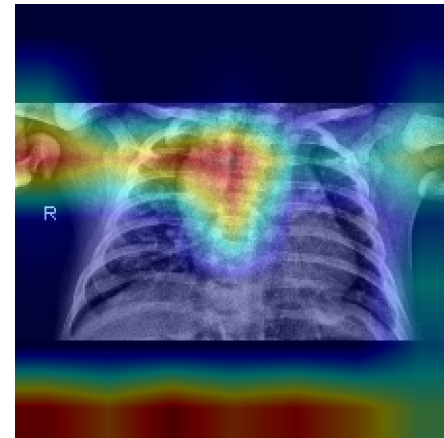
**(b) Preprocessed CLAHE
(Normalized Contrast)**



**(c) Proposed Grad-CAM
(Consolidation Region)**



**(d) Grad-CAM++ Saliency
(Fine Opacity Focus)**



Actual Evaluated Diagnostic Inference:

- Ground-Truth Label: PNEUMONIA (Bacterial Consolidation) | • Predicted Class: PNEUMONIA (True Positive)
- Model Probability: $p = 0.945$ | • Predictive Uncertainty: $\sigma_{mc} = 0.034$ (Low Uncertainty, Confident)